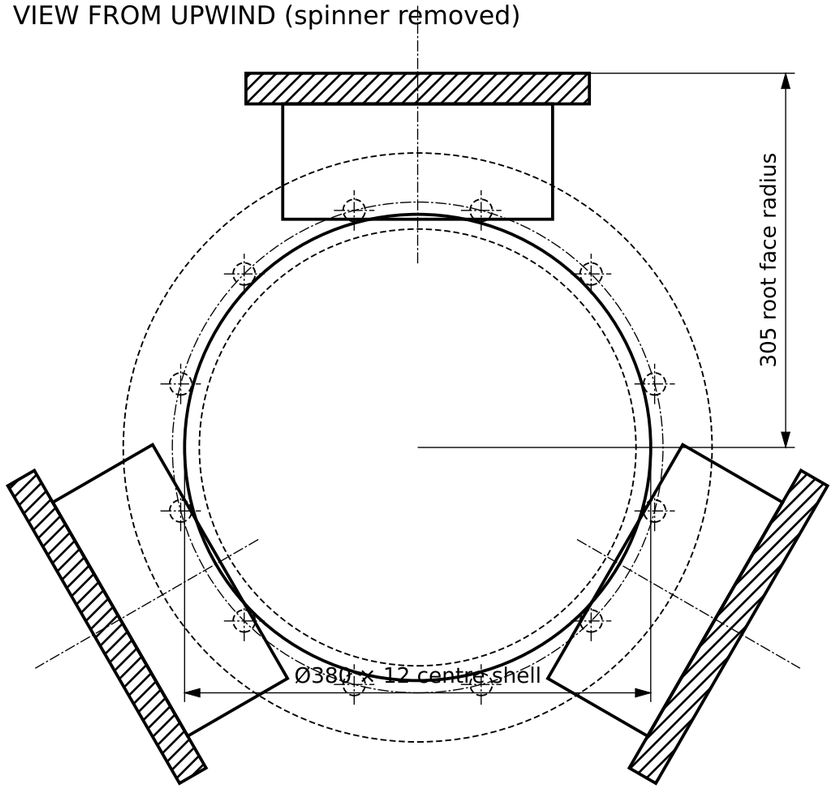
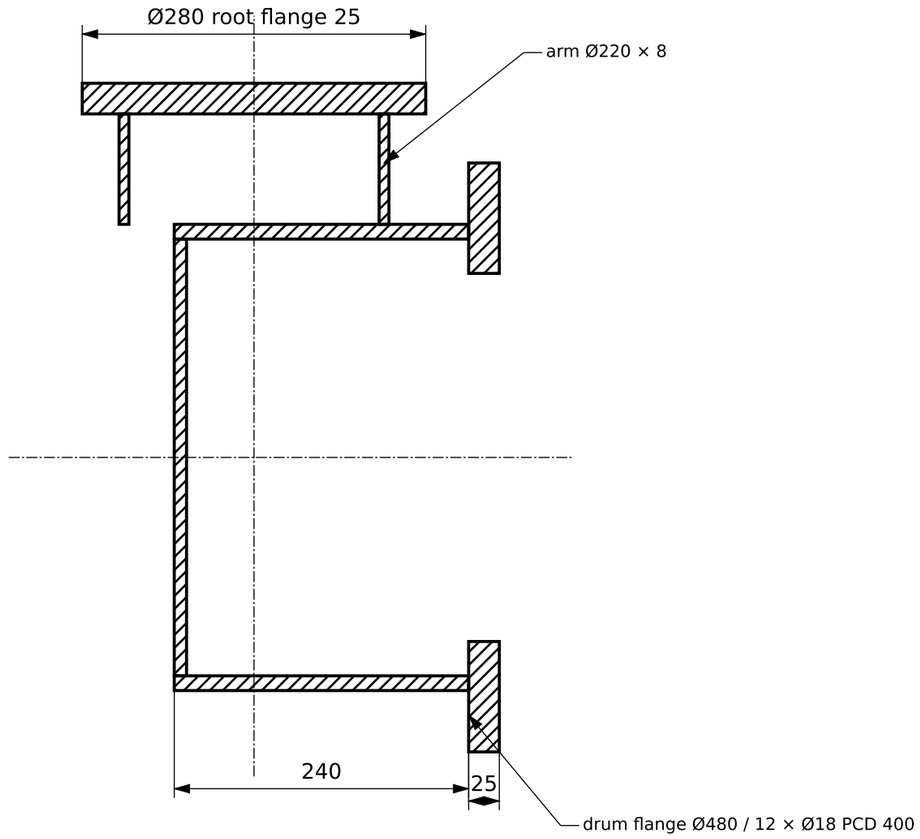
VIEW FROM UPWIND (spinner removed)



SECTION THROUGH THE UPPER ARM



NOTES

1. Arms Ø220 × 8: parked SLM stress 68 MPa (allow 323); fatigue at the flange weld 5.7 MPa with simulated loads (allow 30.8, detail 71).
2. Drum joint: 12 × M16 10.9 PCD 400 (Rev C interface), friction grip, slip OK at 5.7 kNm design torque.
3. Root flange faces machined after stress relief at −2.0° ± 0.05° pitch about each arm axis; 10 × M12 clearance holes Ø13 PCD 185; shim seat for ±2° tapered shims (0.5° steps).
4. Welds full penetration at arm-to-shell and arm-to-flange, ISO 5817 B, 100 % MT + UT on the flange welds. Hot-dip galvanised or zinc sprayed + C5 paint.
5. Front plate Ø356 × 10 closes the shell; spinner brackets (3 × M8) on the arms. FE check of the arm-to-shell weld before release (doc 30 §11).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

FIXED HUB WELDMENT

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-200	A · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355NL (EN 10025-3), welded, stress relieved, machined	87 kg (+ 6 kg spinner)	Claude / not checked